Engineering Applications of Artificial Intelligence 136 (2024) 108972

Contents lists available at ScienceDirect

Engineering Applications of Artificial Intelligence

journal homepage: www.elsevier.com/locate/engappai

Naive Bayes classifier – An ensemble procedure for recall and precision enrichment

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ARTICLE INFO

Keywords:
Naive Bayes
Classification methods
Precision
Recall

ABSTRACT

Data is essential for an organization to develop and make decisions efficiently and effectively. Machine learning classification algorithms are used to categorize observations into classes. The Naive Bayes (NB) classifier is a classification algorithm based on the Bayes theorem and the assumption that all predictors are independent of one another. Since this algorithm is based on probabilities, it is necessary to explore the sample distribution and feature type. This study presents an NB classifier method with enhanced performance among multidimensional and multivariate datasets, named the Naive Bayes Enrichment Method (NBEM). The NBEM is based on automated feature selection using threshold learning and the division of a dataset into sub-datasets according to the feature type. The main advantage of this method is the use of multiple NB classifiers based on different distributions and their combinations to classify a new observation. The final phase includes a weighted classification function that combines the results into a single output. This method was tested with 20 multivariate datasets and compared to other classification models and NB classifier variations. The results showed up to 76.3% improvement in the recall measure using NBEM and up to 43.9% improvement in the F1 score. Furthermore, we found that the error percentage of our method depended on the number of classes.

1. Introduction

In recent years, machine learning has become an increasingly relevant tool in various fields. It can be divided into two broad categories: supervised learning and unsupervised learning (Burkart and Huber, 2021; Hastie et al., 2009; Russell, 2010). For the purpose of the current study, the focus will be on supervised learning. The goal of supervised learning is to instruct the algorithm on what information it should learn to increase its future performance (Sealfon et al., 2021; Sharma et al., 2021). In this learning process, the machine is taught or trained using well-labeled data (i.e., records that have been previously tagged with the correct answers) to predict the label that should be assigned to a new record (Havlíček et al., 2019; Sen et al., 2020). Real-world supervised learning applications include image classification (Chandra and Bedi, 2021), fraud detection (Adewumi and Akinyelu, 2017; Sinayobye et al., 2018), cancer diagnoses (Kourou et al., 2015; Salmi and Rustam, 2019), text classification and analysis (Dai et al., 2007; Xu, 2018), selecting genes with lowest similarity for cancer diagnosis (Azadifar et al., 2022; Rostami et al., 2022), and many others.

Naive Bayes (NB) is a statistical classification technique based on Bayes Theorem (Yang, 2018). It assumes that the effect of a particular feature in a class is independent of other features (Soria et al., 2011). NB is one of the simplest supervised learning algorithms. It calculates the prior probabilities for a given class label and its likelihood probability and returns the conditional probability of a given target (Dai et al., 2007). On the one hand, the NB classifier is much faster than other supervised algorithms as it only calculates the probabilities and has a high-performance rate over categorical data types (Murphy, 2006; Rish, 2001). On the other hand, if the categorical variable includes a category that was not observed in the training set, the model will assign a zero probability and be incapable of making a prediction (Berrar, 2018; Kononenko, 1991; Leung, 2007). This is often known as "zero frequency". To solve this, smoothing techniques, such as Laplace estimation (Boyko and Boksho, 2020; Setyaningsih and Listiowarni, 2021), are used. Another limitation of NB is the assumption of independent predictors, which are highly unlikely to determine a set of independent predictors in the real world.

There are several measures that can be used to evaluate learning algorithms (Demšar, 2006). Classifier accuracy scores are influenced by precision and recall; precision indicates the percentage of correct

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https://doi.org/10.1016/j.engappai.2024.108972
Received 9 November 2022; Received in revised form 12 March 2024; Accepted 10 July 2024
Available online 22 July 2024

predictions and recall indicates how accurate the classifier is from all positive instances. An algorithm with high precision will return more relevant results than those with low precision, while an algorithm with high recall will return the most relevant results (even if irrelevant results are returned as well). In some cases, one metric may have to be prioritized over another. For example, in a spam filter, it may be more necessary to have high precision than high recall (Rusland et al., 2017) since it is more critical to avoid sending significant emails to spam than to catch all emails. The optimal balance of precision and recall depends on the specific application (Winkler et al., 2019).

This study presents a precision and recall enrichment method for NB classifiers, the Naive Bayes Enrichment Method (NBEM). It includes selecting the most significant features using automated threshold learning and a weighted classification function. The procedure divides the dataset into sub-datasets according to the feature types. It then detects and trains the appropriate NB classifier and combines the results using the features' weights. The main advantage of this method is the use of NB classifiers with different distributions and combinations of features to achieve higher classification performances. By using ensemble techniques, the NBEM aims to address the challenges of the "zero frequency" and independence assumptions in NB classifiers. The method can enhance interpretability by revealing how each model contributes to the overall prediction. This can help identify which components increase precision and recall measures and make the classifiers less sensitive to noisy data. Section 2 contains the scientific background and related work. Section 3 will present the procedure, implementation, and correctness of the NBEM. Section 4 will describe the empirical study and a detailed scenario, including different goal demonstrations. The results of our method were compared over six datasets, the results of which will be presented in Section 5. Last, Section 6 will discuss the main conclusions and suggestions for future directions.

2. Related work and state-of-the-art

Automated Machine Learning (AutoML) is emerging together with ML techniques, which allows the automation of processes and tasks required to solve ML problems (Wu et al., 2022; Yao et al., 2018). Several researchers in the field of AutoML have addressed this issue by automatically determining appropriate algorithms and their hyperparameters (He et al., 2021). NB classifiers based on AutoML have been proposed for classifying the content of web pages (Patil and Pawar, 2012), determining an automated sleep score for rats (Rytkönen et al., 2011), generating feature weights automatically (Chen and Wang, 2012), and automating processes in the medical and healthcare fields (Kim and Chen, 2009; Martis et al., 2013; Marucci-Wellman et al., 2015; Svensson et al., 2014).

NB has four main categories (Jiang et al., 2007): Gaussian NB (GNB), used when variables are continuous and assumes that each feature is normally distributed; Categorical NB (CNB), used for categorical features; Multinomial NB (MNB), used to classify discrete (numerical) features; and Bernoulli NB (BNB), used for multivariate Bernoulli models (i.e., Boolean/binary data types). The popularity of this algorithm has made it a significant tool in real-time predictions and dimensionality reduction (Abraham et al., 2007; Chen et al., 2009; Mukherjee and Sharma, 2012; Ratanamahatana and Gunopulos, 2003; Zhang et al., 2009), sentiment analysis (Laksono et al., 2019; Li et al., 2020; Novendri et al., 2020; Ritonga et al., 2021; Villavicencio et al., 2021; Wongkar and Angdresey, 2019), and spam filtering (Agarwal and Kumar, 2018; Ning et al., 2019; Peng et al., 2018). Nevertheless, above and beyond its other uses, NB classifiers have been found to be useful in developing and optimizing recommendation systems (Iwendi et al., 2020; Kulkarni et al., 2018; Narayan and Sathiyamoorthy, 2019).

Table 1
The notation used in this study.

Symbol	Remarks
D	Dataset
m	Number of features
$F = \{F_1,\dots,F_m\}$	Set of features in D
v	Record/vector to classify
$C = \{c_1,\dots,c_k\}$	Set of classes/targets exist in D in increasing order
k	Number of classes/targets that exist in D
$F' \subseteq F$	Subset of significant features by AutoTFS (Koren et al., 2023)
$F_b, F_c, F_g \subseteq F$	Subsets of Boolean, categorical, and numerical features, respectively

In a recent study, NB classifiers were combined with collaborative filtering, a technique in which similar users can filter out items they may be interested in. This method led to a recommendation system that can filter unseen information and predict whether a user is interested in a given resource based on machine learning and data mining techniques (Fayyaz et al., 2020; Valdiviezo-Diaz et al., 2019). For example, it can be applied in the prediction of disease rates (Yu et al., 2019), the analysis and mapping of intelligent cities (Zhang et al., 2022), and the development of general boosting algorithms (Yang et al., 2018). Additionally, NB methodology and classifiers have been applied to estimating the activities of COVID-19 (KC et al., 2021; van de Schoot et al., 2021) and approximating risk probabilities (Ward et al., 2022).

In order to generalize algorithms effectively, studies have recommended ensemble learning, a method that involves learning several algorithms simultaneously with similar objectives (Sagi and Rokach, 2018). Different hyperparameters can be used to perform the same algorithm, or different algorithms can be run and analyzed in various ways (primarily through voting). Therefore, weighting the algorithms together can approximate an almost optimal solution (Dong et al., 2020; Rincy and Gupta, 2020). A significant benefit of this learning method is that it can improve predictions and produce better performance than individual models that contribute to the learning process. Such an example is the Minimum Log-likelihood Difference WNB (MLD-WNB), which optimizes weights according to the Bayes optimal decision rule and provides hyperparameters to control the bias of the model (Ferreira et al., 2001).

Various methods can be used to improve the performance of machine learning algorithms, including feature selection, hyperparameter tuning (i.e., determining the optimal classifier parameters), and others. Selecting the appropriate approach depends on the classifier's behavior. Due to the assumption that all features are independent of NB classifiers, studies suggest improving NB classifiers to address this issue, for example, the Averaged one-dependence estimators (AODE) technique (Webb et al., 2005). As part of the AODE approach, a set of one-dependence estimators (ODEs) is averaged to address the attribute-independence problem. The ODE classifier assumes that each attribute depends on only one other besides the class. Generally, this assumption can provide more accurate results than the NB independence assumption. WAODE, or weightily averaged one-dependence estimators, measures the significance of each input variable and classifies the data by weighting the variable's contribution to the model. A WAODE model yielded significantly better results than a simple AODE model (Jiang and Zhang, 2006). Recent additional studies presented novel methods, including weighting high predictive features to prioritize them over others (Jiang et al., 2016, 2018), adding a hidden variable for each feature describing the influences of all other features, as well as dynamic feature weighting that depends on class (Jiang et al., 2019).

3. Enrichment method for Naive Bayes classifier

This section describes our proposed technique of the NBEM. First, we will detail the three types of NB classifiers used in this study. Then, we will describe the method and its implementation. Last, we will describe the correctness of the method. Table 1 presents the symbols used in this study.

3.1. Naive Bayes classifiers

Given y, the target variable which consists of $\{c_1, \ldots, c_k\}$ classes, this study used three versions of NB classifiers, as follows:

1. **Bernoulli NB** (Manning, 2008) – a classifier for data that is distributed according to the Bernoulli distribution (i.e., binary values). Let v be a vector to classify and $\widetilde{p} = P[v = 1|y]$ the conditional probability of $v = 1$ given the target variable. The decision function is defined as:

$$P[v|y] = \widetilde{p}v + (1 - \widetilde{p})(1 - v)$$

2. **Categorical NB** (Omura et al., 2012) – used for categorically distrusted data. To avoid the "zero frequency" problem, the smoothing parameter ($\alpha > 0$) is used in the decision function as follows:

$$P(v = t|y = c, \alpha) = \frac{N_{vc} + \alpha}{N_c + \alpha n_i}$$

Where N_{vc} is the number of times the category appears in the sample, and n_i is the number of categories that exist in the i^{th} feature.

3. **Gaussian NB** (Zhang, 2005) – the popular and most common classifier of NB. The decision function is derived from the Gauss distribution:

$$P(v|y) = \frac{1}{\sqrt{2\pi\sigma^2}} \cdot e^{-\frac{(v-\mu)^2}{2\sigma^2}}$$

Where μ and σ are the estimated mean and standard deviation using the maximum likelihood principle.

Note. This study was designed to explore the dataset according to the feature types and, therefore, does not use multinomial NB.

3.2. Algorithm

The algorithm inputs a dataset (D) and a vector (v) for classification. First, the AutoTFS procedure (Koren et al., 2023) is used to find F' by means of automated threshold learning. The algorithm uses the output of the AutoTFS procedure and creates $W = \{w_1, \ldots, w_m\}$, a set of weights such that:

$$f_i \in F' \Rightarrow w_{f_i} = \frac{2(|F| - i + 1)}{(2|F| - |F'|)(|F'| + 1)}$$

$$f_i \notin F' \Rightarrow w_{f_i} = \frac{2}{(2|F| - |F'|)(|F'| + 1)}$$

Hence, the significant features are ranked higher than the others, and all features' weights sum up to one (see Section 3.4).

Next, the algorithm uses the weights and creates W_b, W_c, W_g, the sum of Boolean, categorical and numeric features, respectively. The same process creates $\widetilde{D_b}, \widetilde{D_c}, \widetilde{D_g}$, three sub-datasets divided according to their feature type. For each subset, the algorithm adds the target variable and trains the fitted NB classifier, according to the data types. Let $\widehat{y_b}, \widehat{y_c}, \widehat{y_g}$ be the prediction of v by the classifiers. The classification function is defined as:

$$Z(D, v) = (W_b, W_c, W_g) \cdot \begin{bmatrix} \widehat{y_b} \\ \widehat{y_c} \\ \widehat{y_g} \end{bmatrix}$$

In the case of $k = 2$, the dataset has a Boolean target, and the method output is:

$$\widehat{y}(D, v) = \left[\frac{1}{3} \cdot Z(D, v)\right]$$

Otherwise, in cases of $k \geq 3$, the dataset has a multi-class target, and the output is:

$$\widehat{y}(D, v) = \left[\frac{1}{k} \cdot (Z(D, v) - 1)\right]$$

Note that, since the classification method is dependent on k (i.e., the number of classes), the error is controlled by the number of classes. Thus, in the case of $k \geq 3$, small values may cause 'noisy' results. For the justification and analysis, see Section 3.4.

3.3. Implementation

NBEM(D,v, weights**)**
- $F = \{F_1, \ldots, F_m\} \leftarrow$ set of features in D
- $W = \{w_{f_1}, \ldots, w_{f_m}\} \leftarrow$ set of weights by AutoTFS
- $A = \{\widetilde{D_b}, \widetilde{D_c}, \widetilde{D_g}\} \leftarrow$ Boolean, categorical, and numerical datasets, respectively
- If weights are used:
  - o $W_b, W_c, W_g \leftarrow \sum_{f_i \in D_b} w_{f_i}, \sum_{f_i \in D_c} w_{f_i}, \sum_{f_i \in D_g} w_{f_i}$
- Else, set each W_b, W_c, W_g by uniform distribution.
- For each $D_i \in A$:
  - o Add the target variable to $\widetilde{D_i}$
  - o Train NB classifier by the data type
- $\widehat{y_b}, \widehat{y_c}, \widehat{y_g} \leftarrow$ Prediction of v by the Boolean, categorical, and numerical classifiers
- Return $\widehat{y}(D, v)$

3.4. Correctness

Let $F = \{F_1, \ldots, F_m\}$ be a set of features in dataset D, and $C = \{c_1, \ldots, c_k\}$ be a set of classes/targets (encoded string to index) existing in D in increasing order (i.e., $c_1 \le c_2 \le \ldots \le c_k$). The AutoTFS procedure creates F', a subset of significant features. F' is a sorted subset, such that the first feature is ranked as the most significant feature. Therefore, the weights are assigned according to the following:

$$f_i \in F' \Rightarrow w_{f_i} = |F| - i + 1$$
$$f_i \notin F' \Rightarrow w_{f_i} = 1$$

Hence, the most significant features are ranked higher than the others. To ensure standardization, the total weights are bounded by one. The sum of the weights of all features in F' is:

$$\sum_{f_i \in F'} w_{f_i} = |F| + (|F| - 1) + \ldots + (|F| - |F'| + 1)$$

This is an arithmetic series with $|F'|$ items. Thus, the total sum is:

$$\sum_{f_i \in F'} w_{f_i} = \frac{|F'|}{2}(|F| + |F| - |F'| + 1) = \frac{1}{2} \cdot |F'| \cdot (2|F| - |F'| + 1)$$

The same process is calculated for $f \notin F'$:

$$\sum_{f_i \notin F'} w_{f_i} = 1 + 1 + \ldots + 1 = |F| - |F'|$$

They are then combined to find the sum of the total weights:

$$\sum_{f_i \in F'} w_{f_i} + \sum_{f_i \notin F'} w_{f_i} == \frac{|F'|}{2}(2|F| - |F'| + 1) + |F| - |F'| == \frac{|F'|(2|F| - |F'| + 1) + 2(|F| - |F'|)}{2} =$$
$$= \frac{2|F||F'| - |F'|^2 + |F'| + 2|F| - 2|F'|}{2} =$$
$$= \frac{2|F||F'| - |F'|^2 - |F'| + 2|F|}{2} =$$
$$= \frac{2|F|(|F'| + 1) - |F'|(|F'| + 1)}{2} =$$
$$= \frac{(2|F| - |F'|)(|F'| + 1)}{2}$$

Each item is divided by the total sum to standardize the weights for a sum up to one:

$$f_i \in F' \Rightarrow w_{f_i} = \frac{2(|F| - i + 1)}{(2|F| - |F'|)(|F'| + 1)}$$

$$f_i \notin F' \Rightarrow w_{f_i} = \frac{2}{(2|F| - |F'|)(|F'| + 1)}$$

Let v be a vector to classify and let $\widetilde{D_b}, \widetilde{D_c}, \widetilde{D_g}$ be Boolean, categorical, and numerical datasets, respectively. The values of W_b, W_c, W_g are defined as:

$$W_b = \sum_{f_i \in D_b} w_{f_i}$$
$$W_c = \sum_{f_i \in D_c} w_{f_i}$$
$$W_g = \sum_{f_i \in D_g} w_{f_i}$$

Since $\widetilde{D_b}, \widetilde{D_c}, \widetilde{D_g}$ have no common features (i.e., no feature exists in more than one dataset), and by the fact that all weights sum up to one, it holds that $0 \le W_b, W_c, W_g \le 1$. Once the NB classifiers have completed the training phase, the values $\widehat{y_b}, \widehat{y_c}, \widehat{y_g}$ represent the classification of the Boolean, categorical, and numerical classifiers, respectively. The classification function is divided into two cases: $k = 2$ (i.e., Boolean target) or $k \ge 3$ (i.e., multi-class target).

Case $\boldsymbol{k} = \mathbf{2}$

The values of $\widehat{y_b}, \widehat{y_c}, \widehat{y_g}$ are either zero or one, thus:

$$0 \le \widehat{y_b}, \widehat{y_c}, \widehat{y_g} \le 1$$

By multiplying the above equations, we get that for each $i \in \{b, c, g\}$ it holds that $W_i \widehat{y_i}$ is bounded between zero and one. Therefore, the total sum of the three equations is:

$$0 \le W_b \cdot \widehat{y_b} + W_c \cdot \widehat{y_c} + W_g \cdot \widehat{y_g} \le 3$$

To achieve an output within the range of the given classes, the equation is divided by three to obtain the classification function for Boolean targets:

$$\widehat{y} = ROUND\left(\frac{1}{3} \cdot (W_b, W_c, W_g) \cdot \begin{bmatrix} \widehat{y_b} \\ \widehat{y_c} \\ \widehat{y_g} \end{bmatrix}\right)$$

The values of $\widehat{y_b}, \widehat{y_c}, \widehat{y_g}$ are bounded between c_1 to c_k (i.e., the output of the NB classifiers must recommend one of the existing classes). Combine it with the weights' equations to obtain:

$$\forall\, i \in \{b, f, g\} : c_1 \le \widehat{y_i} \le c_k$$

$$\forall\, i \in \{b, f, g\} : 0 \le W_i \widehat{y_i} \le c_k$$

To sum up all of the equations:

$$0 \le W_b \cdot \widehat{y_b} + W_c \cdot \widehat{y_c} + W_g \cdot \widehat{y_g} \le 3c_k$$

Using algebraic operations, minus one was applied and then divided in k:

$$-1 \le W_b \cdot \widehat{y_b} + W_c \cdot \widehat{y_c} + W_g \cdot \widehat{y_g} - 1 \le 3c_k - 1$$

$$-\frac{1}{k} \le \frac{1}{k}\left(W_b \cdot \widehat{y_b} + W_c \cdot \widehat{y_c} + W_g \cdot \widehat{y_g} - 1\right) \le \frac{3c_k - 1}{k}$$

For every $k \ge 3$, it holds that:

$$\frac{3c_k - 1}{k} \le \frac{kc_k - 1}{k}$$

Thus:

$$-\frac{1}{k} \leq \frac{1}{k}\left(W_b \cdot \hat{y_b} + W_c \cdot \hat{y_c} + W_g \cdot \hat{y_g} - 1\right) \leq \frac{kc_k - 1}{k}$$

$$-\frac{1}{k} \leq \frac{1}{k}\left(W_b \cdot \hat{y_b} + W_c \cdot \hat{y_c} + W_g \cdot \hat{y_g} - 1\right) \leq c_k - \frac{1}{k}$$

$$\hat{y} = ROUND\left(\frac{1}{k} \cdot \left(\left(W_b, W_c, W_g\right) \cdot \begin{bmatrix} \hat{y_b} \\ \hat{y_c} \\ \hat{y_g} \end{bmatrix} - 1\right)\right)$$

Since the algorithm rounds the result, and each $k \geq 3$, the value of $c_k - \frac{1}{k}$ is rounded to c_k and the prediction is bounded by c_k.

3.5. Computational complexity

Let m be the number of features, n be the number of observations, and $C = \{c_1, \ldots, c_k\}$ be the set of classes in dataset D. In the brute force approach, each NB classifier requires $O(n \cdot m \cdot k)$ operations. By pre-processing, it can be optimized to $O(m \cdot k)$ operations. Let $W = \{w_{f_1}, \ldots, w_{f_m}\}$ be the set of weights of each feature. The proposed method divides the weights according to the type of variable, which can be done in one iteration over W and, therefore, requires $O(m)$ operations. Last, the classification function multiplies two vectors of size three (i.e., constant size) and, thus, requires a constant time of $O(1)$. The total complexity time of the proposed method is:

$$O(m \cdot k) + O(m) + O(1) = O(m(k+1))$$

In comparison, other ensemble methods, such as bagging or boosting, typically entail a higher training complexity than Naive Bayes due to the need to train multiple base models. For instance, bagging trains each base model according to a random subset of the data, leading to a complexity of $O(B \cdot n \cdot m \cdot k)$, where B is the number of base models. Similarly, boosting trains base models sequentially, where each subsequent model focuses on the data points misclassified by the previous models, resulting in a complexity that can be exponential, in the worst case. The proposed NBEM method retains the favorable computational efficiency of the basic Naive Bayes classifier. Therefore, the total time complexity of the proposed method is $O(m(k+1))$, which is comparable to the optimized complexity of a single Naive Bayes classifier.

4. Empirical study

4.1. Data Sources

The following datasets were compared to examine the NBEM method.

1. **Diabetes** (Strack et al., 2014). A dataset of 100,000 diabetic and non-diabetic patients. It consists of 54 medical features of mixed types. The target variable indicates whether the patient's health has improved after diabetes medications.
2. **Fetal Health** (Ayres-de-Campos et al., 2000). An infant and maternal health classification dataset. A total of 2126 observations are spread across 21 features, along with a single target variable with three possible values: normal, suspect, and pathological.
3. **Heart Disease** (Dua and Graff, 2019). This database contains 76 features and 368 observations. The target variable refers to the presence of heart disease in the patient, ranging between 0 and 4.
4. **Students' Academic Success** (Realinho et al., 2021). A dataset of information about undergraduate students from multiple higher education institutions. Thirty-six features are provided related to 4424 students' enrollment and academic performance. The target variable has three options: graduate, dropout, or enrolled.
5. **Bank** (Moro et al., 2014). Campaigns by Portuguese banks to market directly to consumers. With over 45,210 observations, the dataset contains seven numerical (continuous) and nine categorical features. The target variable indicates "1″ if the client subscribed to a term deposit and "0″ otherwise.
6. **Heart Failure** (Detrano et al., 1989). This dataset consists of 12 clinical characteristics of 299 heart failure patients, and a Boolean target variable that indicates whether the patient had heart failure.
7. **Internet Advertisements** (Kushmerick, 1998). A set of advertisements on the internet containing a binary target variable (Kushmerick, 1998); 1557 features were recorded, with over 3279 observations in the dataset.
8. **Chronic Kidney Disease** (Rubini and Eswaran, 2015). A total of 400 patients were studied over two months, with 24 features, including personal and medical data. An indication of chronic kidney disease was accompanied by a Boolean target variable.
9. **Mushroom (1987)**. This dataset includes descriptions of gilled mushrooms. This dataset contains 8124 observations involving 22 features, as well as a Boolean target variable that specifies whether a mushroom is edible (or poisonous).
10. **Dry Beans** (Koklu and Ozkan, 2020). A total of 13,611 images of dry beans were analyzed, consisting of 16 features and a target variable of seven bean classes.
11. **Abalone** (Nash et al., 1995). Using 4177 observations and eight features, this dataset aims to predict the age of abalone according to physical measurements.
12. **Car Evaluation** (Bohanec, 1997). This dataset includes six features about cars (e.g., doors, safety, etc.) with a total of 1728 samples.
13. **Credit Approval** (Quinlan, n.d.). Credit card applications are described in this dataset, with over 15 mixed type features and 690 observations.
14. **E. coli** (Nakai, 1996). This dataset comprises protein localization with 336 E. coli proteins split into eight classes.
15. **Glass Identification** (German, 1987). A dataset of six types of glass with nine multivariate features.
16. **Hepatitis (1988)**. This dataset includes information about 155 patients with Hepatitis, with over 19 multivariate features.
17. **Productivity Prediction** Imran et al. (2021). This dataset includes 14 attributes of the garment manufacturing process and employee productivity. A total of 1197 records were collected manually and validated by industry experts.
18. **Haberman Survival** (Haberman, 1999). Data was collected from 1958 to 1970, including details on survival after breast cancer surgery. The dataset and study were conducted at the University of Chicago's Billings Hospital.
19. **Indian Liver Patients** (Ramana and Venkateswarlu, 2012). The dataset comprises 584 patient records, with over 11 features, collected from Andhra Pradesh, India. The data is based on information about several biochemical markers to determine whether a patient suffers from liver disease.
20. **Cirrhosis Survival** (Dickson et al., 2023). Data utilizing 17 clinical features of 418 patients to predict the survival state of patients with liver cirrhosis.

4.2. Experimental procedure

For each dataset described in Section 4.1, the following experiment was performed.

1. Before beginning each experiment, the datasets were cleaned of missing values. Note that the original dataset was used, and we did not implement a normalization method.
2. For AutoTFS (Koren et al., 2023), we set $T = 0.2$, which compared different thresholds between 0 and 0.2 with steps of 0.01; $\alpha = 4$ was

used, as in the original paper. For example, if $T = 0.1$ and $\alpha = 2$, then all selected features must have at least 10% importance for the model and be selected by at least two of the different techniques.
3. For the classification measures comparison, we used the k-fold cross-validation with a default value of $k = 10$ (King et al., 2021).
4. We compared the results of our method with the following models and hyperparameters:
   a. Random Forest (RF) Classifier with a total of 200 estimators and Shannon information gain as splitting criteria.
   b. Gradient Boosting Machines (GBM) Classifier with a learning rate of 0.1 and log-loss as optimization function.
   c. AdaBoost Classifier with a default learning rate of 1.0 and a total of 200 estimators.
   d. Weighted Average One-Dependence Estimators (WAODE).
   e. Gaussian Naïve Bayes.
   f. Minimum Log-likelihood Difference Weighted Naïve Bayes (MLD-WNB).
5. For each model described in 4.2.4, we compared the precision, recall, and overall accuracy.

4.3. Use case: the diabetes dataset

This section presents a complete example of the NBEM method over the diabetes dataset. For simplicity of explanation, the comparison is made between our method NBEM and Gaussian NB. A broader comparison, including additional classification models and variations of NB classifiers, can be found in Section 5.

The diabetes dataset consisted of 54 features and a target variable with over 100,000 observations. Each observation represented a diabetes patient. The target variable was a Boolean value, with "1" representing a change in the diabetes level after taking medications, or "0" otherwise (i.e., $C = \{0, 1\}$ and $|C| = k$). After removing empty features and values, the dataset included 45 features and 96,452 observations.

Of the 45 features, there were 9 Boolean (denoted as F_b), 23 categorical (denoted as F_c), and 13 numerical features (denoted as F_g). Using the AutoTFS procedure (Koren et al., 2023), weights were assigned to the features. Given that W_b, W_c, W_g were the sum of the weights of the Boolean, categorical, and numerical features, respectively, we received $W_b = 0.170, W_c = 0.560, W_g = 0.27$.

Next, the algorithm divided the dataset into three subsets: $\widetilde{D_b}, \widetilde{D_c}, \widetilde{D_g}$ including the features in F_b, F_c, F_g, respectively. For each dataset, the algorithm created and trained the NB classifier according to the feature type. Given that $\widehat{y_b}, \widehat{y_c}, \widehat{y_g}$ represents the classification of the Boolean, categorical, and numerical classifiers, respectively, the classification function was updated as follows:

$$\frac{1}{3}\left(0.170 \cdot \widehat{y_b} + 0.560 \cdot \widehat{y_c} + 0.270 \cdot \widehat{y_g}\right)$$

For further analysis, we used the original dataset and compared the results of the Gaussian NB to the other NB classifier results. However, in this dataset, we could not compare the Bernoulli and categorical NB classifiers as they contradict the assumption of the variable types.

Models need high recall when the goal is to build output-sensitive predictions. For example, predicting cancer or diabetes requires a high recall to cover the false negative results. Table 2 compares the precision (classifier exactness: percentage of correct predictions from all instances classified correctly), recall (classifier completeness: percentage of correct predictions from all positive), and overall accuracy between the Gaussian NB to the NBEM.

Table 2
Accuracy, precision, and recall comparison between Gaussian NB and NBEM.

		Gaussian NB	NBEM	Improvement
Accuracy		0.751	0.970	29.16%
Precision	**Class 0**	0.710	0.961	35.35%
	Class 1	0.850	0.992	16.70%
Recall	**Class 0**	0.911	0.990	8.67%
	Class 1	0.570	0.958	68.07%

Table 3
A comparison between the confusion matrix of the Gaussian NB and MBEM classifiers.

		Predicted	
		Negative (0)	Positive (1)
Actual	**Negative (0)**	48.8(***53.2***)	4.7(***0.5***)
	Positive (1)	20.2(***2.4***)	26.3(***43.9***)

Note. The data is presented as A(***B***), where A is the percentage achieved by Gaussian NB and ***B*** is the percentage acquired by the NBEM.

The NBEM improved the overall accuracy by 29.16%, from 0.751 to 0.970. While accuracy is the ratio of the successful performance of the classifier, precision and recall contribute to the understanding of how it functions and if there are any "struggles" among the classifiers. A significant high improvement was shown in class "1" (i.e., diabetic patients), with a recall of 68.07%. Given that TP and FN are the true positive and false negative values, the recall was calculated by:

$$Recall = \frac{TP}{TP + FN}$$

Any recall improvement requires minimizing the FN to achieve a recall close to one. As mentioned earlier, there are cases in which the goal is to build a classifier with a sensitive prediction, such as in the medical field. Due to its notable improvement, we examined and compared the confusion matrices of both classifiers (Table 3). The Gaussian NB achieved 20.2% FN, while our method reduced it to 2.4%. As this dataset included information on diabetic and non-diabetic patients, our method successfully reduced the number of patients who were found negative as "non-diabetic" but who were actually diabetic.

Area Under Receiver Operating Characteristic Curve (AUC-ROC) is another way to assess the quality of a classifier. Essentially, it separates the 'signal' from the 'noise' by plotting the TP rate against the false positive (FP) rate under various threshold values. AUC measures a classifier's ability to differentiate between classes and is calculated from the ROC curve. The higher the AUC, the better the classifier performs. Fig. 1 illustrates the influence of the true negative (TN) and TP values on the ROC. The figure on the right presents the distributions of the TN and TP, and the figure on the left presents the ROC. The model has an ideal separability measure when the two distribution curves do not overlap. When two distributions overlap (i.e., the FN and FP groups are created), the ROC value is reduced, and the curve may be linear (i.e., random classifier without any pattern).

In the diabetes dataset, the original AUC-ROC score was 0.739, while our method yielded a score of 0.969, representing a 31.12% improvement. We can conclude that our method resulted in more accurate estimates than the original model of patients with diabetes. Fig. 2 presents the ROC of the Gaussian NB and the NBEM.

Fig. 2 shows that the ROC curve created by our procedure was sharper, while the Gaussian NB curve was closer to a random classifier. Thus, enhancing the precision and recall values may be an improvement over enhancing the overall accuracy, especially in datasets where each record is significant for the result, as in medical datasets.

5. Results

Ten datasets were compared to assess and evaluate the NBEM results. For comparison, we used the F-β score, which combines the precision and recall by calculating the harmonic sum. The β parameter represents the desired goal: the case of $\beta < 1$ is used when the FP has more impact than the FN, otherwise, $\beta > 1$. Here, we used $\beta = 1$ (i.e., the F1 score), which is used when FN and FP have equal importance for the result. Table 4 compares the accuracy and the F1 score of each dataset (all are

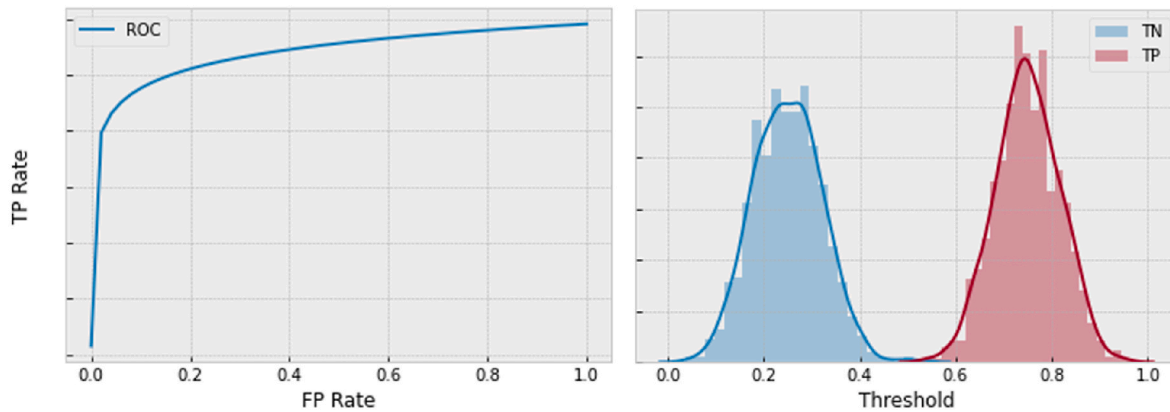

Fig. 1. Illustration of the impact of TN and TP distributions on the RUC.

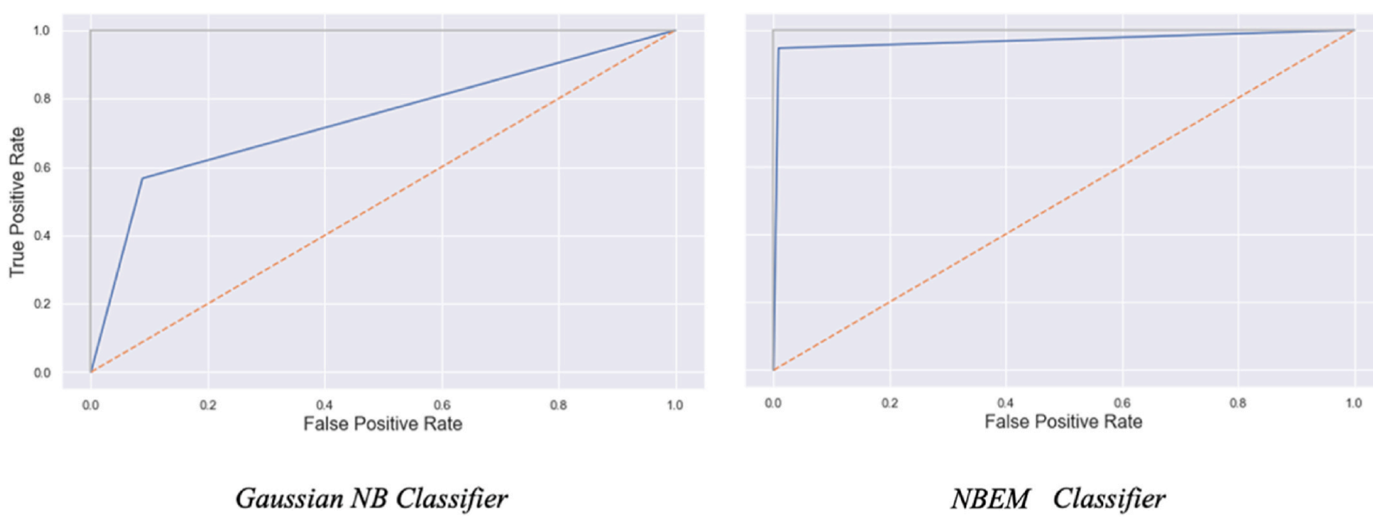

Fig. 2. ROC comparison between Gaussian NB and the NBEM.
Note. The blue line represents the ROC curves. The orange dashed line represents a random classifier result.

Table 4
A comparison of the F1-score between NBEM and other classification models, including variations of the NB classifier.

	NBEM	GNB	RF	GBM	AdaBoost	WAODE	WNB
Diabetes	0.971	0.742	0.867	0.968	0.979	0.903	0.735
Fetal Health	0.862	0.768	0.802	0.786	0.793	0.860	0.833
Heart Disease	0.905	0.805	0.888	0.898	0.878	0.851	0.872
Students	0.883	0.808	0.833	0.783	0.725	0.615	0.711
Bank	0.901	0.829	0.884	0.866	0.882	0.881	0.794
Heart Failure	0.763	0.706	0.736	0.701	0.657	0.616	0.693
Internet Ads	0.954	0.838	0.922	0.917	0.895	0.930	0.706
Kidney Disease	0.899	0.799	0.892	0.887	0.856	0.878	0.845
Mushroom	0.621	0.527	0.586	0.549	0.444	0.641	0.612
Dry Beans	0.937	0.759	0.926	0.923	0.832	0.802	0.811
Abalone	0.798	0.701	0.768	0.733	0.761	0.720	0.655
Car Evaluation	0.919	0.697	0.919	0.754	0.787	0.801	0.835
Credit Approval	0.863	0.814	0.865	0.847	0.817	0.823	0.712
E. coli	0.912	0.761	0.904	0.761	0.742	0.733	0.693
Glass Identification	0.801	0.621	0.796	0.759	0.681	0.699	0.659
Hepatitis	0.944	0.802	0.901	0.945	0.912	0.899	0.886
Productivity Prediction	0.962	0.903	0.959	0.909	0.919	0.894	0.913
Haberman Survival	0.898	0.796	0.895	0.884	0.890	0.852	0.855
Indian Liver Patients	0.698	0.613	0.648	0.689	0.662	0.641	0.655
Cirrhosis Survival	0.711	0.679	0.705	0.699	0.678	0.631	0.615

mixed type datasets) between our method and classification models described in 4.2.4.

We conducted a Friedman test to determine statistically significant performance differences between the models (Appendix 1). For a p-value of 0.05, the results indicated a statistically significant performance difference between at least some of the models. The Nemenyi test was

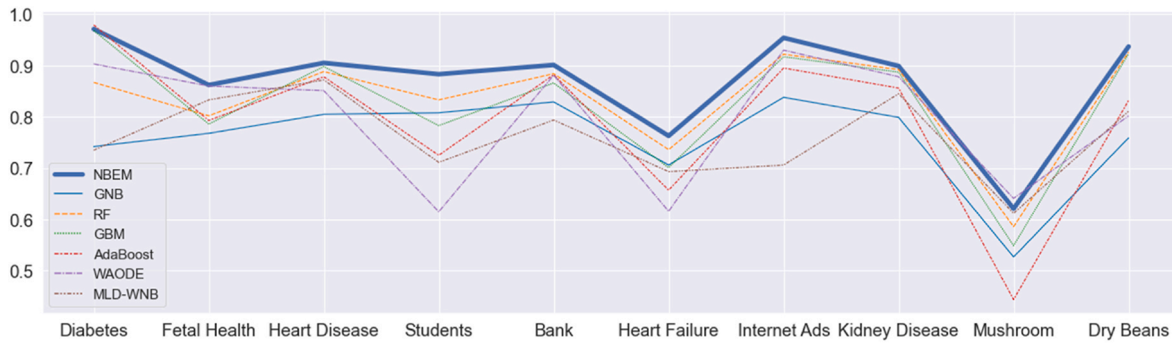

Fig. 3. Average F1-score comparison between NBEM and other classification models.

employed following the Friedman test for a more detailed analysis. This post-hoc test aided in identifying the specific pairs of models that exhibited statistically significant differences in performance. The Nemenyi test indicated that our model statistically outperformed all other models except the Random Forest model. The GNB, GBM, AdaBoost, WAODE, and WNB had p-values of less than 0.05, demonstrating the proposed model's strong performance compared to the other algorithms. Only the Random Forest model comparison yielded a p-value higher than 0.05. The results suggest that the NBEM and the Random Forest models have similar performance capabilities or that the datasets used influenced the results.

An improvement was shown for most of the datasets in F1-score measures using the proposed method. For the other cases, there was no significant difference between the result produced by the NBEM method and the optimal result. For example, in the diabetes dataset, NBEM achieved an F1-score of 0.971, whereas AdaBoost achieved an F1-score of 0.979. As described in Section 3.4, the number of classes controlled the error classification range. Furthermore, some datasets had high percentages of improvements, such as the diabetes and heart failure datasets, and others showed slight improvements (see Table 4). Further examination of our classifier completeness (i.e., recall) is presented in Fig. 3, which compares the F1-score of each dataset achieved by our method and the compared classification models.

As presented in Fig. 3, the NBEM method (i.e., the bold blue line) scored a higher F1-score than all of the compared datasets. Since we used $\beta = 1$, which is used when FN and FP have equal importance for the result measure, we conclude that our method succeeded in enhancing both precision and recall compared to all other classification models and

Table 6
An ablation study of NBEM classifier over the diabetes dataset.

	Precision	Recall
NBEM Classifier	0.982	0.961
(−) Feature weights	0.933	0.904
(−) Bernoulli NB	0.975	0.951
(−) Categorical NB	0.863	0.833
(−) Gaussian NB	0.911	0.907

NB classifier variations. Beyond the fact that all compared datasets had higher recall with our method, there were datasets in which its improvement was significant while others showed a slight improvement, such as the fetal health dataset. As previously mentioned, as it was a medical dataset (classifying diseases in fetuses), any minor change could be significant. In a closer analysis of the improvement percentage in this dataset (1.04%), it can be concluded that the NBEM was able to "correct" the classification of between 5 and 6 additional observations, given that we used 25% of the dataset for the test set and the overall dataset has 2216 observations.

As explained in Section 3, the classification error is a function of the number of existing classes/targets (k). Table 5 compares the precision and recall achieved by our method to several classification models. Our method balanced precision and recall compared to other classification models that, in most cases, harmed one index to enhance the other. For example, in the "internet advertisements" dataset, our method achieved a precision of 0.96 and a recall of 0.95, while RF presented a precision of 0.94 and a recall of 0.89. Similarly, the WNB over this dataset raised a

Table 5
Precision and recall comparison between NBEM and other classification models, including variations of NB classifier.

	NBEM	GNB	RF	GBM	AdaBoost	WAODE	WNB
Diabetes	0.98(0.96)	0.76(0.72)	0.82(0.88)	0.96(0.96)	0.95(0.98)	0.91(0.89)	0.74(0.73)
Fetal Health	0.86(0.88)	0.77(0.74)	0.79(0.82)	0.76(0.77)	0.81(0.79)	0.85(0.88)	0.83(0.83)
Heart Disease	0.91(0.90)	0.82(0.80)	0.89(0.86)	0.90(0.85)	0.88(0.84)	0.86(0.84)	0.87(0.88)
Students	0.89(0.86)	0.79(0.82)	0.83(0.83)	0.76(0.82)	0.73(0.69)	0.62(0.60)	0.75(0.68)
Bank	0.91(0.90)	0.84(0.81)	0.89(0.86)	0.86(0.87)	0.89(0.86)	0.88(0.86)	0.80(0.77)
Heart Failure	0.76(0.75)	0.71(0.66)	0.72(0.75)	0.71(0.68)	0.67(0.62)	0.61(0.61)	0.71(0.69)
Internet Ads	0.96(0.95)	0.85(0.82)	0.94(0.89)	0.91(0.91)	0.88(0.93)	0.93(0.94)	0.73(0.65)
Kidney Disease	0.91(0.88)	0.82(0.77)	0.90(0.86)	0.89(0.85)	0.85(0.84)	0.87(0.86)	0.83(0.86)
Mushroom	0.62(0.64)	0.54(0.51)	0.59(0.56)	0.56(0.51)	0.44(0.46)	0.64(0.62)	0.61(0.62)
Dry Beans	0.94(0.91)	0.77(0.74)	0.93(0.90)	0.93(0.90)	0.86(0.79)	0.81(0.77)	0.83(0.79)
Abalone	0.79(0.81)	0.70(0.71)	0.74(0.79)	0.71(0.77)	0.76(0.77)	0.70(0.73)	0.67(0.63)
Car Evaluation	0.93(0.90)	0.69(0.71)	0.93(0.90)	0.78(0.73)	0.78(0.78)	0.76(0.83)	0.82(0.84)
Credit Approval	0.87(0.86)	0.79(0.82)	0.86(0.87)	0.82(0.86)	0.84(0.77)	0.83(0.81)	0.71(0.71)
E. coli	0.91(0.93)	0.76(0.76)	0.92(0.90)	0.75(0.78)	0.73(0.77)	0.76(0.72)	0.67(0.72)
Glass Identification	0.79(0.82)	0.63(0.60)	0.81(0.77)	0.76(0.74)	0.65(0.69)	0.71(0.68)	0.69(0.63)
Hepatitis	0.95(0.94)	0.82(0.79)	0.92(0.90)	0.94(0.95)	0.91(0.92)	0.88(0.91)	0.88(0.89)
Productivity Prediction	0.95(0.96)	0.91(0.89)	0.96(0.95)	0.92(0.89)	0.92(0.91)	0.88(0.91)	0.93(0.88)
Haberman Survival	0.89(0.91)	0.77(0.82)	0.89(0.90)	0.89(0.86)	0.88(0.89)	0.87(0.84)	0.86(0.83)
Indian Liver Patients	0.71(0.68)	0.62(0.61)	0.67(0.63)	0.67(0.72)	0.65(0.69)	0.67(0.62)	0.64(0.66)
Cirrhosis Survival	0.72(0.71)	0.69(0.66)	0.71(0.70)	0.70(0.68)	0.66(0.69)	0.65(0.63)	0.61(0.63)

Note. The data is represented as Precision(Recall).

precision of 0.73 and recall of 0.65, which is lower and noisier than other models.

An ablation study is conducted to investigate the interaction between NBEM classifier components. We conducted an ablation study to explore these interactions, meaning we applied the techniques simultaneously, but switched one off at a time as each technique was applied. The ablation study is shown in Table 6, alongside the accuracy results. The first row in Table 6 presents the NBEM classifier precision and recall, while all other rows describe the changes in the precision and recall achieved by removing each component of the NBEM classifier.

As presented in Table 6, removing the Categorical NB classifier from the NBEM method yielded the highest differences in the precision and recall measures. Thus, it has the highest impact and effectiveness among NBEM components. The Bernoulli NB scored as the least significant component of this dataset. It may be due to the number of binary features (seven) compared to categorical (23) and numerical features (11).

6. Discussion and conclusions

This study proposed a new method, the Naive Bayes Enrichment Method, based on Naive Bayes classifiers for precision and recall enrichment for multivariate datasets. The NBEM utilizes threshold learning for automatic feature selection and subdivides a dataset based on the feature type. Its main advantage is using multiple NB classifiers based on different distributions and types and their combinations to classify new observations. The NBEM aims to address the challenges posed by the NB classifiers' zero frequency and independence assumptions. On the one hand, using ensemble techniques, the NBEM classifier can enhance performance by handling sensitive and noisy data. On the other hand, ablation studies can identify which components of the NBEM classifier lead to improved precision and recall. We tested our method over 10 multivariate and mixed target (binary and multi-class) datasets. Here are the main conclusions.

1. A weighted combination of different NB classifiers shows an increase in the precision and recall measures and, consequently, in the overall accuracy. For example, the diabetes dataset presents an improvement of 25.6% in precision using our method compared to the Gaussian NB. Considering the recall, the heart failure dataset showed an improvement of 76.3%, from 0.456 using the Gaussian NB to 0.804 using the NBEM.
2. With the help of the ablation study, we found that without the feature weights component, the classifier precision was reduced by 4.98% and the recall by 5.93%. The most significant component of the NBEM procedure over the diabetes dataset is the categorical NB classifier, which yields a decrease of 12.11% in precision and 13.31% in the recall. Therefore, it can be concluded that the distribution of weights is essential in our classifier. Also, this conclusion may stem from the number of features of each type.
3. To understand the mutual effect of both precision and recall, we calculated the F1 score (i.e., its harmonic sum) and compared the Gaussian NB and the NBEM. The improvements in the F1 score were more moderate, since it is an average of both measures, and showed improvements between 1.6% (in the fetal health dataset) and 43.9% (in the heart disease dataset).
4. Using Friedman and Nemenyi statistical tests, it was determined that the proposed NBEM model outperformed most of the other models (i.e., GNB, GBM, AdaBoost, WAODE, WNB). The only model that the NBEM did not outperform was the Random Forest model, which appeared to have similar performance. Additional data or a different dataset may be necessary to determine whether there is a clear difference in the current model's results compared to the Random Forest model.
5. While deep learning models demonstrate remarkable accuracy and can be used instead of NB classifiers, they often function as "black boxes" where the decision-making process is less transparent. This can be a concern in medical settings where it is crucial to understand the reasoning behind a diagnosis or treatment recommendation. Naïve Bayes classifiers, while sometimes slightly less accurate, offer greater interpretability, allowing healthcare providers to see which factors contribute most to a model's prediction.

Several factors could affect the results, including the quality of the data, the number of features, and the organization's goals. An incorrect adjustment of parameters or a lack of data could also cause this method to fail. Future work should examine the generalization of the classification function (described in Section 2) into a single case fit to all dataset types (binary or multi-class). In addition, future work should address and investigate the following issues.

1. Compare different values of k (i.e., the number of classes/targets), which might present more profound results on the influence of k on the classification function and its performance. Such a comparison may yield conclusions on the precision and recall improvement rates as a function of k.
2. Examine the NBEM classifier according to the distribution of the features and not necessarily according to their type. For example, the NBEM can be expanded using the Multinomial NB and additional distribution measures.
3. Design a preprocessing procedure for feature selection and reduce the overall correlation among the features. Since the input is a multivariate dataset, it requires sophisticated feature selection combining different correlation (or dependency) measures.

Role of the funding source

This research did not receive any specific grants from funding agencies in the public, commercial, or not-for-profit sectors.

Data statement

The data used in this study is available and accessible in UCI Machine Learning Repository – Datasets (https://archive.ics.uci.edu/ml/datasets.php).

CRediT authorship contribution statement

Or Peretz: Data curation, Software, Visualization. **Michal Koren:** Formal analysis, Methodology, Project administration, Writing – original draft, Writing – review & editing. **Oded Koren:** Conceptualization, Methodology, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data used in this study is available and accessible in UCI Machine Learning Repository – Datasets (https://archive.ics.uci.edu/ml/datasets.php).

Appendix 1

The Friedman test was used to identify performance differences between the various algorithms. The following table presents the rankings of the F1-scores obtained from Table 4. For each algorithm, the average ranking was calculated over 20 datasets, denoted as $\bar{r}_j$.

	NBEM	GNB	RF	GBM	AdaBoost	WAODE	WNB
Diabetes	2	6	5	3	1	4	7
Fetal Health	1	7	4	6	5	2	3
Heart Disease	1	7	3	2	4	6	5
Students	1	3	2	4	5	7	6
Bank	1	6	2	5	3	4	7
Heart Failure	1	3	2	4	6	7	5
Internet Ads	1	6	3	4	5	2	7
Kidney Disease	1	7	2	3	5	4	6
Mushroom	2	6	4	5	7	1	3
Dry Beans	1	7	2	3	4	6	5
Abalone	1	6	2	4	3	5	7
Car Evaluation	1	7	1	6	5	4	3
Credit Approval	2	6	1	3	5	4	7
E. coli	1	3	2	3	5	6	7
Glass Identification	1	7	2	3	5	4	6
Hepatitis	2	7	4	1	3	5	6
Productivity Prediction	1	6	2	5	3	7	4
Haberman Survival	1	7	2	4	3	6	5
Indian Liver Patients	1	7	5	2	3	6	4
Cirrhosis Survival	1	4	2	3	5	6	7
Average Ranking ($\bar{r}_j$)	1.20	5.90	2.60	3.65	4.25	4.80	5.50

Evaluating the Friedman test requires finding the test statistics with $N = 20$ and $k = 7$:

$$Q = \frac{12N}{k(k+1)} \sum_{j=1}^{k} \left(\bar{r}_j - \frac{k+1}{2}\right)^2 = \frac{12 \cdot 20}{7 \cdot 8} \cdot \sum_{j=1}^{7} (\bar{r}_j - 4)^2 = 263.76$$

With a p-value <0.05, it can be concluded that there was a statistically significant performance difference between at least some of the models.

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